

SEEDLINE · REQUIREMENTS

SeedLine requirements

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DOCUMENT CONTROL

Document ID	SDL-REQ-001	Version	0.2
Status	Draft	Date	2026-09-25
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Project TRL	TRL 2 Technology concept formulated		
Repository	github.com/BoujeeEnjinia1701/seedline		
Source	docs/03-requirements.md @ 5a78719		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with concept status against each
0.1	2026-09-24	Amish Chadha	Initial scaffold

SeedLine requirements

These are first-pass requirements for the concept. Targets are proposals for review, not yet validated with users, and will be checked by calculation at TRL 3 and revised after co-design sessions (see SDL-PRB-001).

The **design case** is maize at 250 mm in-row spacing on 0.75 m rows, sown 50 mm deep in a tilled loam seedbed at a walking speed of 3 km/h (about 0.83 m/s). The **small-seed case** is carrot or lettuce seed of 1 to 2 mm at 25 to 50 mm spacing on a prepared vegetable bed.

■ **Table 1. Requirements**

ID	Requirement	Target	Verification (TRL 3 or later)	Concept status (estimate)
R1	Single-seed placement, design case	Misses 5 % or less and multiples 5 % or less of cells, in the terms of ISO 7256-1	Metering calculation; later bench test on a sticky belt	About 5 % misses, 4 % multiples: met on paper, no margin
R2	Spacing uniformity	Coefficient of variation of in-row spacing 30 % or less in the design case	Later field count	Unverified; depends on seed bounce in the drop tube
R3	Spacing range	25 to 400 mm in-row spacing, set by plate cell count and one sprocket change	Spacing formula (SDL-PRC-001)	About 22 to 390 mm: not met above 390 mm
R4	Spacing follows travel	Seed spacing set by distance travelled, not speed; wheel slip 8 % or less	Slip estimate; later field test	3 to 8 % slip estimated: met, thin margin
R5	Seed size range	Seeds from 1.5 to 15 mm in their largest dimension with plates of the same diameter	Plate design review; later bench test	At risk below about 2 mm (FDM cell accuracy)
R6	Printable plates	Each plate prints in 2 h or less on a 180 x 180 mm bed in PLA, PETG or ASA with no supports; cell size within ± 0.2 mm	Slicer estimate; later measurement	About 1 h, 25 g PETG: met on paper; accuracy unverified
R7	Crop change	Plate swapped in 2 min or less by hand, with no tools, without emptying more than the seed in the housing	Design review; later timed trial	Side door and hand knob proposed: met on paper
R8	Sowing depth	Adjustable 10 to 60 mm in 10 mm steps, held within ± 10 mm on a tilled seedbed	Design review; later field check	Slotted shank bracket: met on paper; control on rough seedbeds unverified
R9	Work rate	0.1 ha/h or more in the design case, including turns and refills	Field capacity estimate	About 0.14 ha/h: met
R10	Push effort	Horizontal push force 150 N or less in the design case	Force estimate; later spring-scale test	About 100 N on a good seedbed, up to about 220 N on a heavy one: at risk
R11	Handling	Mass 15 kg or less; handle grip height adjustable from 850 to 1,050 mm; lifted and turned at a row end by one person	Mass estimate; later weighing	About 14 kg: met, thin margin
R12	Hopper	2 L or more, fills from a jug without spilling, empties in 1 min or less when changing crops	Model volume	About 2.1 L: met
R13	Row spacing kit	Marker sets the next row at 200 to 900 mm from the current row, within ± 25 mm	Design review	Telescoping arm: met on paper
R14	Guarding	Chain and sprocket nip points covered on the side facing the operator's hands and feet; no exposed sharp edges above the opener	Design review	Band guard in the model: met on paper
R15	Durability	Frame, drive and opener last 5 seasons of about 2 ha each with only chain, brush and		Unverified

ID	Requirement	Target	Verification (TRL 3 or later)	Concept status (estimate)
		plate replacement; plates last 1 season or more	Design review; later wear test	
R16	Prototype cost	Prototype parts within the \$400 budget	Priced BOM (bom/ bom.csv)	About \$223: met
R17	Replication cost	Base seeder (all items except the row marker kit) about \$200 or less in parts	Priced BOM	About \$199: met, no margin
R18	Local build	Frame, handle and opener made from common steel sections with a drill and a welder or bolts; drive from standard #35 roller chain or bicycle parts	Design review	Met on paper

■ Requirements not met or at risk

- **R3 not met:** the longest spacing is about 390 mm with three cells and the smallest wheel sprocket. Wider spacings would need a plate with fewer cells, a 10 T sprocket or a skip-cell plate. Proposed: relax the target to 390 mm or add a skip plate, awaiting Amish.
- **R5 at risk:** cells for 1 to 2 mm seed are near the accuracy of a 0.4 mm FDM nozzle. Pelleted seed, a 0.25 mm nozzle or resin-printed plates may be needed.
- **R10 at risk:** the force estimate is about 100 N on a good seedbed but reaches about 220 N on a heavy or cloddy one, mostly from opener draft and from the downward share of the push along a steep handle.
- **R1, R4 and R11** are met only with thin or no margin.

■ Assumptions

- Ground wheel effective diameter 300 mm (942 mm travel per turn) before slip.
- Walking speed 3 km/h; field efficiency 65 % for turns, refills and stops.
- Soil rolling resistance coefficient 0.2 to 0.3 on tilled soil for narrow wheels; opener draft 40 to 80 N at 50 mm depth. Both are estimates to check at TRL 3.
- Miss and multiple rates follow typical figures for cell-type mechanical meters with a brush; they are estimates, not measured values for this plate.